where VWAP is the volume weighted average price over the trading period. A positive value indicates better performance and a negative value indicates underperformance.

Interval VWAP comparison serves as a good measure of execution quality and does a nice job of accounting for actual market conditions, trading activity, and market movement. The interval VWAP, however, does suffer from three issues. First, the larger the order the closer the results will be to the VWAP price, as the order price will become the VWAP price. Second, actual performance can become skewed if there are large block trades that occur at extreme prices (highs or lows) in crossing venues, especially in cases where investors have limited opportunity to participate with those trades. Third, the VWAP measure does not allow easy comparison across stocks or across the same stock on different days. For example, it is not possible to determine if missing VWAP by 3 bps in one stock is better performance than missing VWAP by 10 bps in another stock. If the first stock has very low volatility and the second stock has very high volatility, missing VWAP by 10 bps in the second name may in fact be better performance than missing VWAP by 3 bps in the first name.

There are three different VWAP performance metrics used: full day, interval, and VWAP to end of day.

Full Day VWAP: Used for investors who traded over the entire trading day from open to close. There is currently no "official" VWAP price on the day but many different providers, such as Bloomberg, Reuters, etc., do offer one. These vendors determine exactly what trades will be included in the VWAP calculations but they may not use all the market trades. For example, some providers may filter trades that were delayed or negotiated because they do not feel these prices are indicative of what all market participants had fair access to.

Interval VWAP: Used as a proxy for the fair market price during the time the investor was in the market trading. The interval VWAP is a specific VWAP price for the investor over their specific trading horizon and needs to be computed from tic data. This is in comparison to a full day VWAP price that is published by many vendors.

VWAP to End of Day: Used to evaluate those orders that were completed before the end of the day. In these cases, the broker or trader made a conscious decision to finish the trade before the end of the day. This VWAP to End of Day provides some insight into what the fair market price was including even after the order was completed. It helps determine if the decision to finish the order early was appropriate. This is a very